

Informatics 141

Computer Science 121

Information Retrieval

Lecture 4

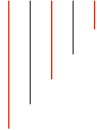
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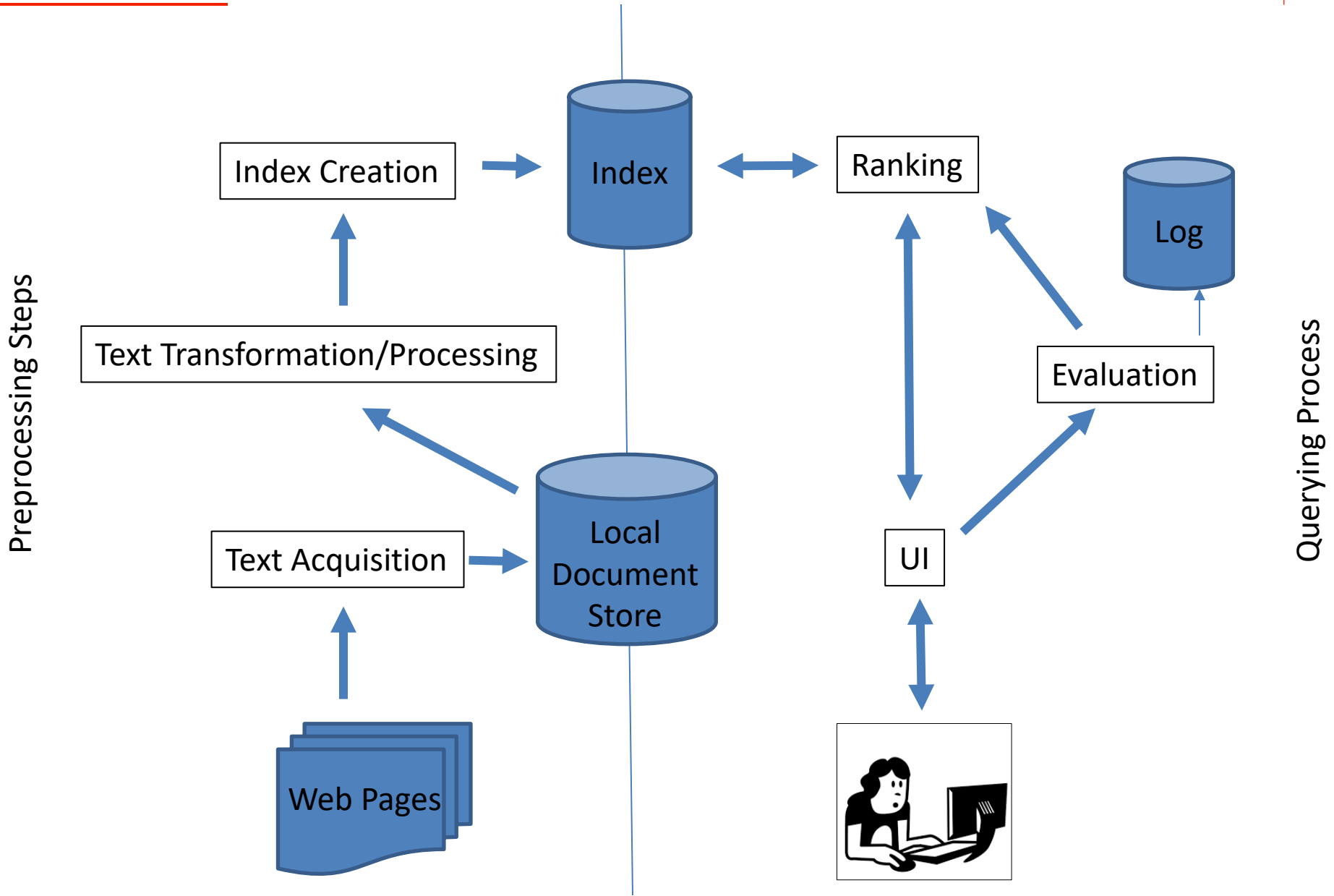
Tokenization

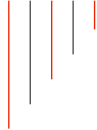
Part 1

Information Retrieval

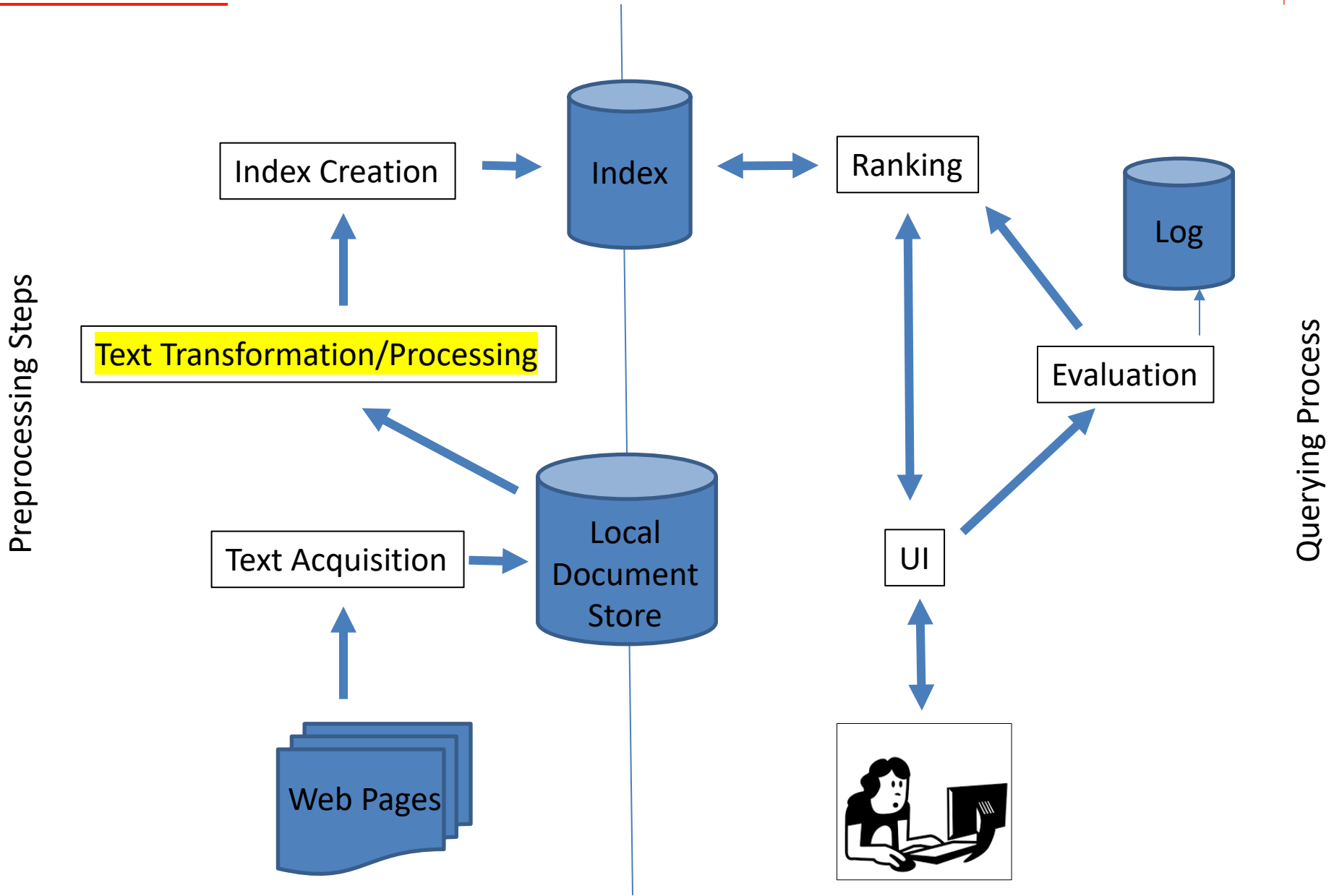


Architecture : Tokenization?

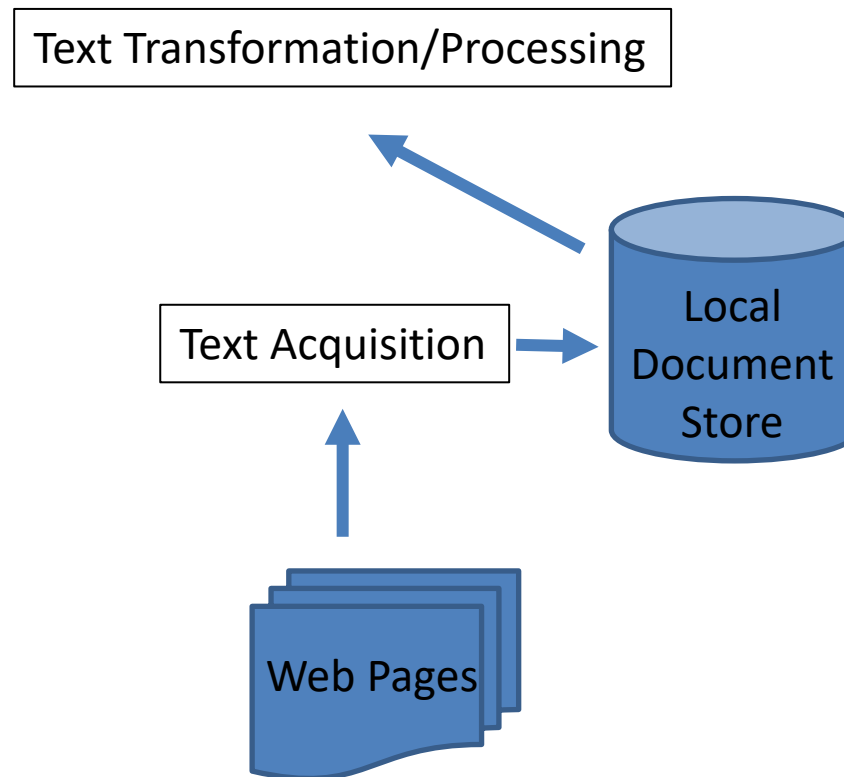




Architecture : Text Transformation and Processing



- The complete set of documents that are available to be searched



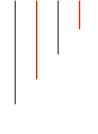


- The complete set of documents that are available to be searched
- In a websearch context:
 - The full set of pages that will be considered by the search engine

Text Processing



- The step/stage of processing the documents/pages in your corpus for use in your search engine
- Occurs prior to building your index to improve the efficiency/performance



- Typically 2 main steps in text processing:
 - **Tokenization (lexical analysis)**
 - Breaking the text into tokens
 - **Linguistic pre-processing**
 - Applying rules to the tokens to improve the efficiency of the index

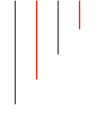
Tokenization

- **What is tokenization?**



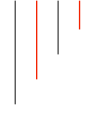
- Break the input into simple units of meaning
 - Can be a word or not! Depends on the choice of how to create tokens...
 - Character stream -> token stream
 - Called a tokenizer / lexer / scanner
- Compiler front-end
 - Tokenizer / lexer / scanner may hook up to a parser
- Preprocessor for information retrieval
 - Tokenizer / lexer / scanner feeds tokens to retrieval system

Identifying Tokens



- Divide on whitespace and throw away punctuation?
- What is a token? Depends...
 - Apostrophes
 - O'Neill
 - Aren't
- Hyphen-handling
 - clear-headed vs clearheaded
 - mother-in-law

Identifying Tokens



- Multiple words as single token?
 - San Francisco
 - white space
 - New York University vs York University
- Tokens that aren't words...
 - info@uci.edu
 - http://www.ics.uci.edu/
 - 192.168.0.1

Identifying Tokens

- **Early** tokenization methodology:
 - A sequence of 3 or more alphanumeric characters
 - A space or special character indicates the end of the token
 - All characters were converted to lower case

Identifying Tokens

- Early tokenization methodology:
 - Example:

“Bigcorp's 2007 bi-annual report showed profits rose 10%.”

Identifying Tokens

- Early tokenization methodology:
 - Example:

“Bigcorp's 2007 bi-annual report showed profits rose 10%.”



“bigcorp 2007 annual report showed profits rose”

- Issues:
 - Too generic and simple : *resulting in a lot of lost information*

Assignment 1 : Tokenizer from scratch

You should write the tokenizer in Python
(3.10).

This will help you with your next two assignments!

Assignment 1 : Tokenizer from scratch

Very important: **At certain points, the assignment may seem underspecified – this is by design.**

In those cases, make your own choices and assumptions and be prepared to defend them.

Assignment 1 : Tokenizer from scratch

Your program must run!

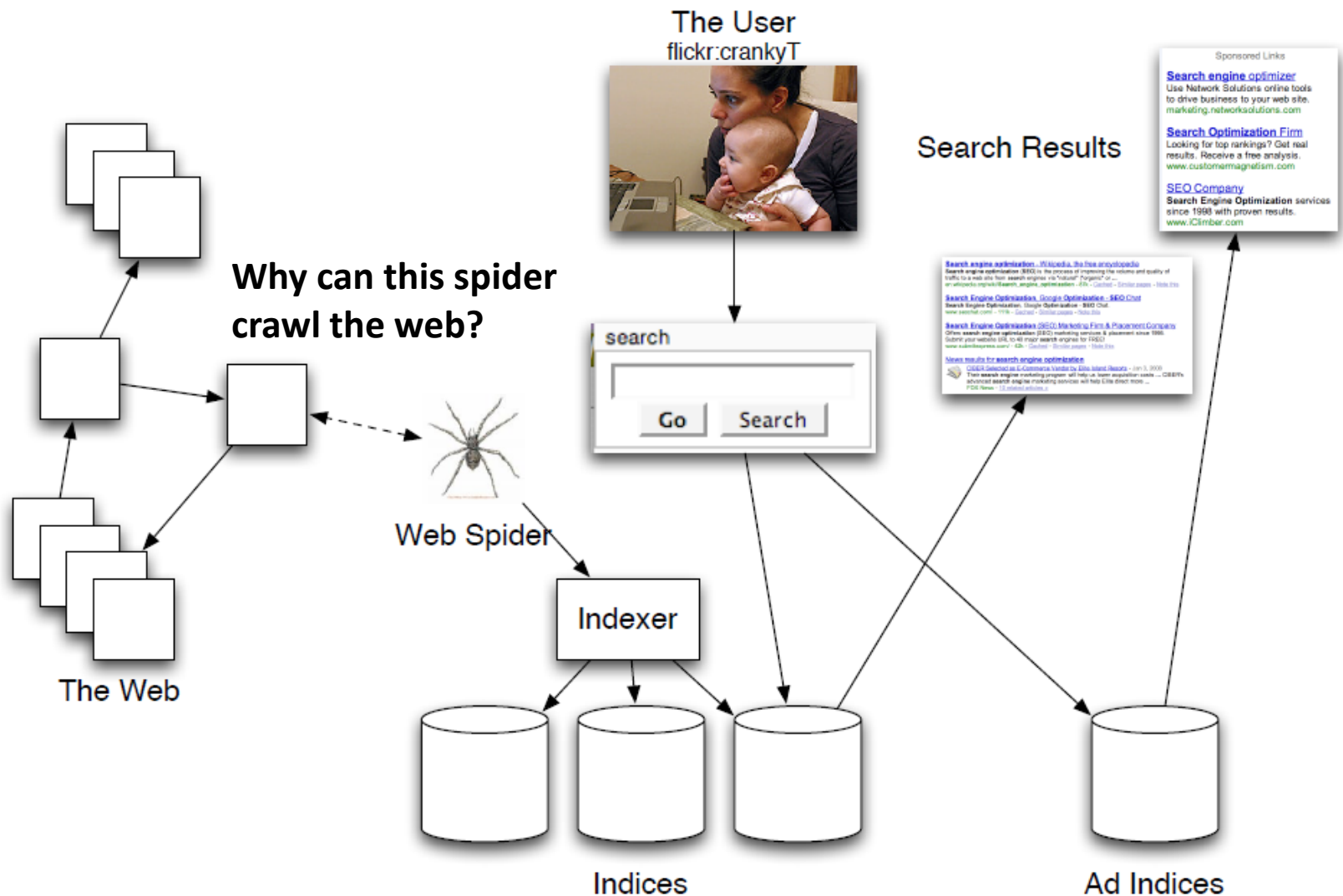
It must be executable from the command line.

You should get the file names from command line arguments.

A bit further on the web

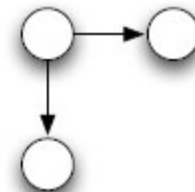
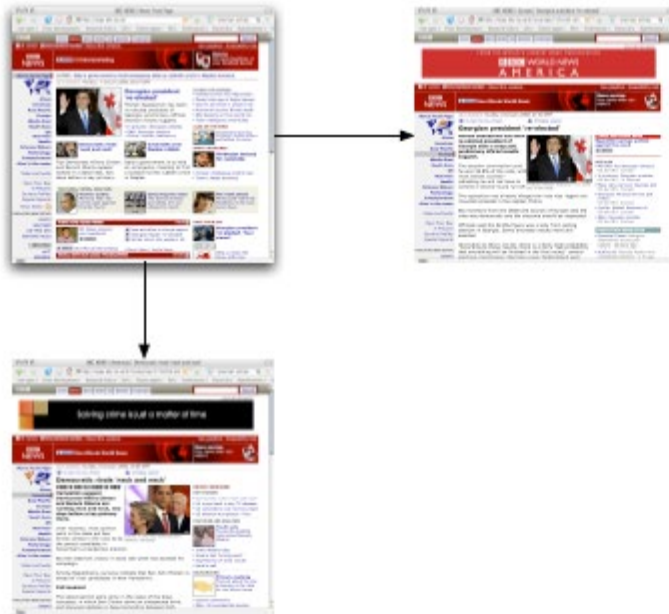
Information Retrieval

Web Search Engine



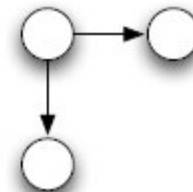
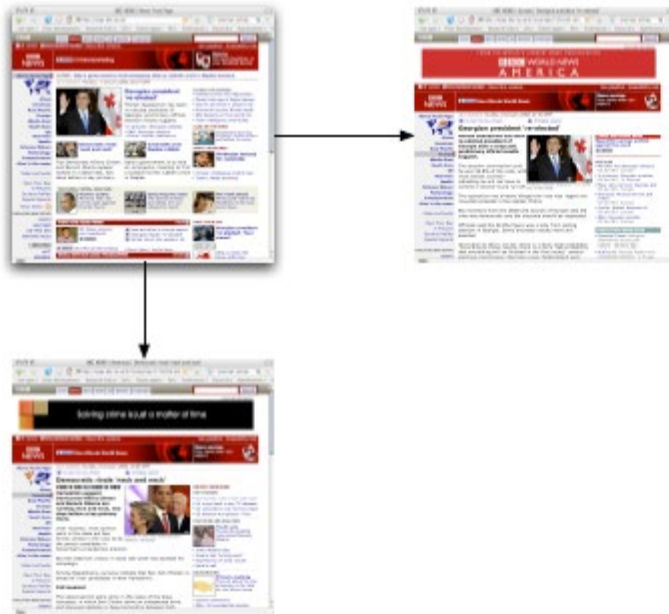
The Web Graph

- The Web is a graph
 - Pages are nodes
 - Hyperlinks are directed edges



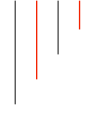
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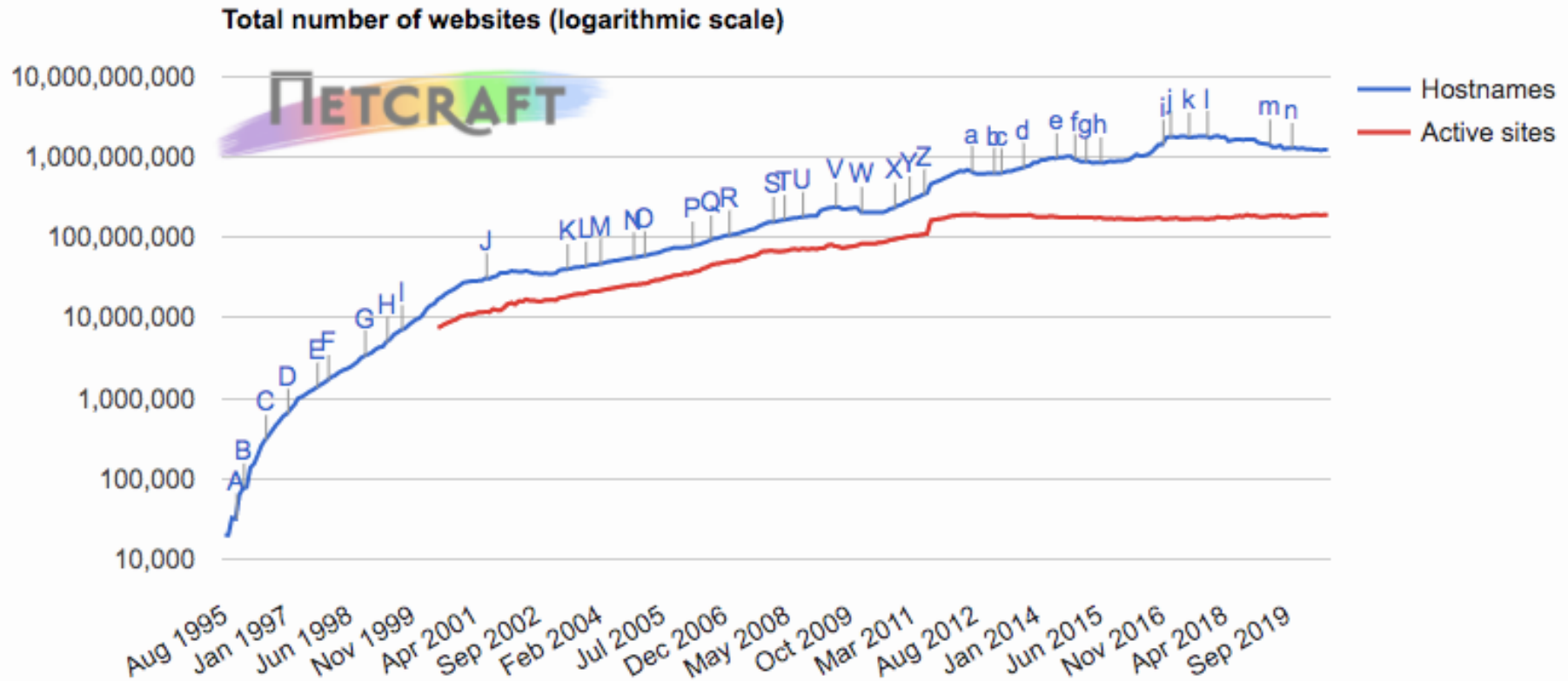
What are the characteristics of this graph?

Characteristics



- Significant duplication
 - 30%-40% in some studies [Brod97, Shiv99]
 - 25%-30% Google's number from 2013
[<https://www.youtube.com/watch?v=mQZY7EmjbMA>]
 - www.copyscape.com
- High linkage
 - More than 8 links per page
- Very large
 - *How large?*

Web Characteristics: Size



<https://news.netcraft.com/archives/2020/09/23/september-2020-web-server-survey.html>

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 - *Hosts* $O(10^9)$
 - *Pages* $O(???)$
- Spam, misleading and false information
 - $O(10^{8-9?})$ of pages of it

Characteristics

- High rate of change
 - [Cho00] 720k pages from 270 popular sites sample daily for 5 months in 1999
 - ?? % changed weekly, ?? % daily

How much do you think?

Characteristics

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 - [Fett02] 151M pages checked over a few months
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Characteristics



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 - [Fett02] 151M pages checked over a few months
 - **Significant changes: 7% weekly**
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 - [Ntul04] 154 **large** sites recrawled from scratch weekly
 - **8% had new pages every week**
 - **8% die**
 - **5% new content**
 - **25% new links**

The Web: Evolution

- 1st phase: static content (documents)
- 2nd phase: dynamic content (applications)
- 3rd phase: user-generated content (social)
- 4th phase: mobile
- 5th phase: ??? AR? VR? Brain-interfaces?



WIKIPEDIA
The Free Encyclopedia

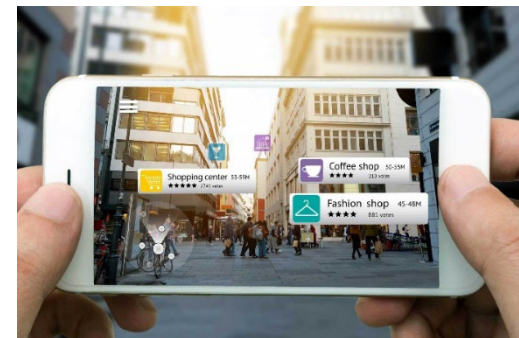


The Web: Evolution

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- 3rd phase: user-generated content (social)
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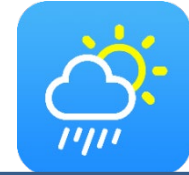
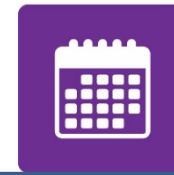


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The Web: Evolution

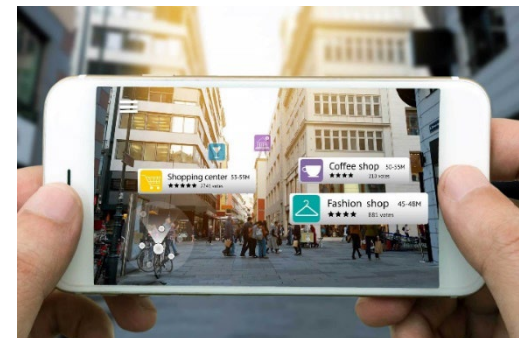
- 1st phase: static content (documents)
- 2nd phase: dynamic content (applications)



One need has been kept invariant: to find information

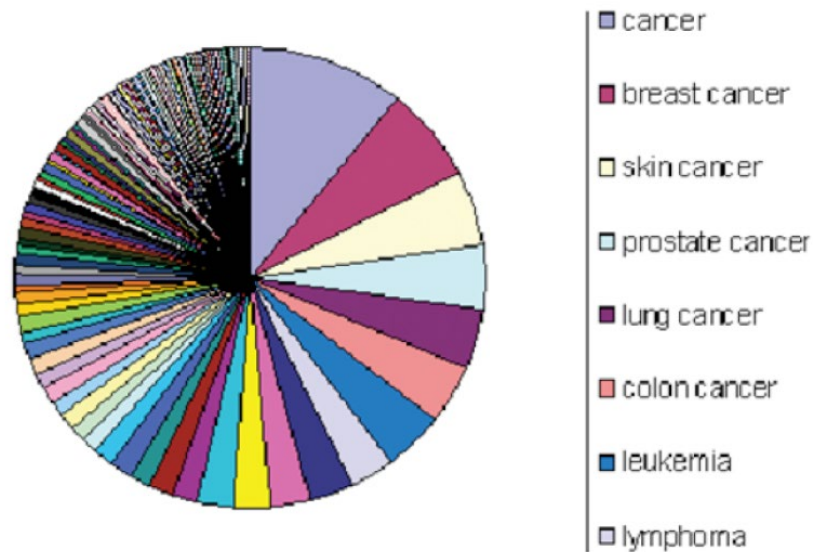
Present-day solution : IR Systems for SEARCH and RANKING

- 4th phase: mobile
- 5th phase: AI agents



The Web: Web search characteristics

- Few popular broad queries
- Many rare specific queries
 - E.g. distribution of all cancer related searches:

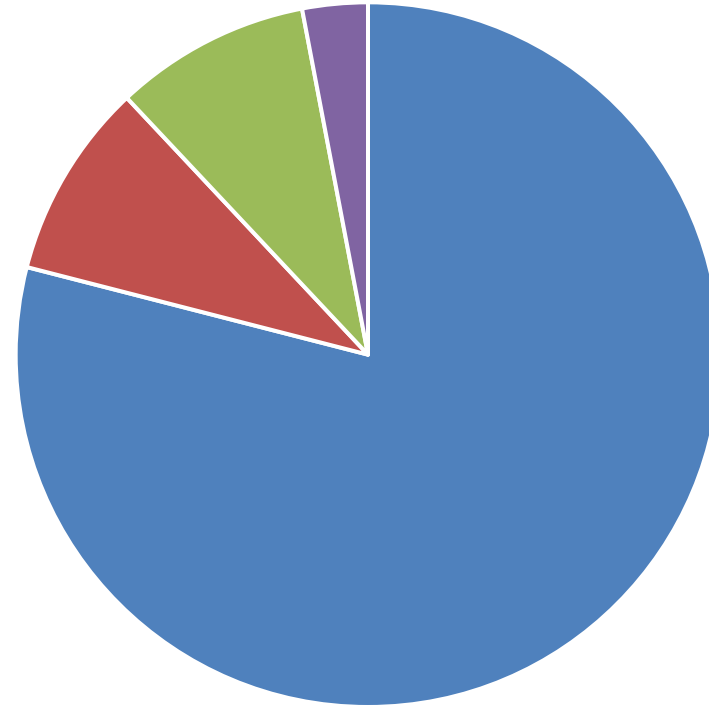


Web Users: Popular Searches

- Term Interest can be explored at google trends
 - <https://www.google.com/trends/>
- A search engine **uses trend information** to
 - Auto-complete:
 - Enhance user experience;
 - Correct spelling errors;
 - Optimize search results
 - Trending “near you” can help in relevance scoring;
 - Smart-cache search results
 - If some search is trending, keep results in cache!

Web Users: Motivations

- [Jansen et al. 2008]
 - Informational needs (~80%)
 - Want to learn about something
 - Navigational needs (~10%)
 - Want to go to that page
 - Transactional needs (~10%)
 - Want to do something
 - Miscellaneous
 - Exploration, social, etc



■ Information ■ URL Search ■ Transaction ■ Others

Web Users: Query formation

- Most queries are **ill-defined** queries

Web Users: Query formation

- Most queries are **ill-defined** queries
 - Do you have any idea of the size of queries?

Web Users: Query formation

- Most queries are **ill-defined** queries
 - Short (80% < 3 words)
 - Imprecise terms
 - No logical operators

Web Users: Query formation

- Most queries are **ill-defined** queries
 - Short (80% < 3 words)
 - Imprecise terms
 - No logical operators
 - Low effort: spelling mistakes

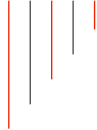
Web Users: Characteristics

- Wide variance in
 - Needs
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Your search engine will be better if you take these points seriously

Web Users: Characteristics

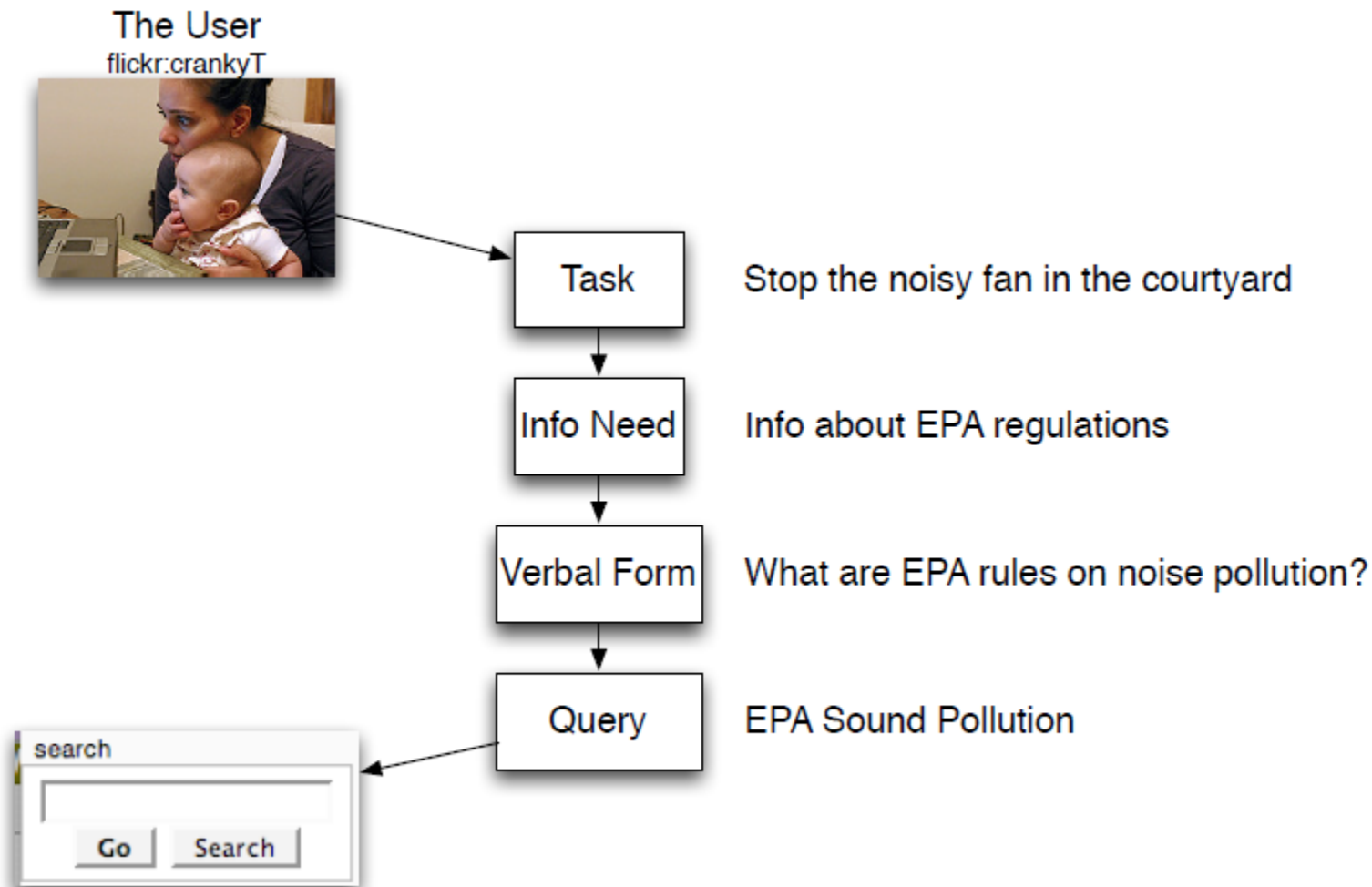
- Wide variance in
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- Behavior
 - 85% look over one result screen only (!!!)

Web Users: Characteristics



- Wide variance in
 - Needs
 - Expectations
 - Knowledge
 - Bandwidth
- Behavior
 - 85% look over one result screen only (!!!)
 - 78% of queries are not modified
 - Follow links (“The scent of information”)

Information need pipeline



Answering the need behind the query

- Query is often imprecise indicator of what the user wants
- What can we do to improve this?

Answering the need behind the query

- Query is often imprecise indicator of what the user wants
- What can we do to improve this?
 - Design retrieval models and mathematical definitions for the **Relevance Score** that are generic, albeit not ideal for all possible “specific topics”.
 - User context: who, where, what
 - Guess the type of information (image, map, math, etc.)
 - Correct and/or expand queries